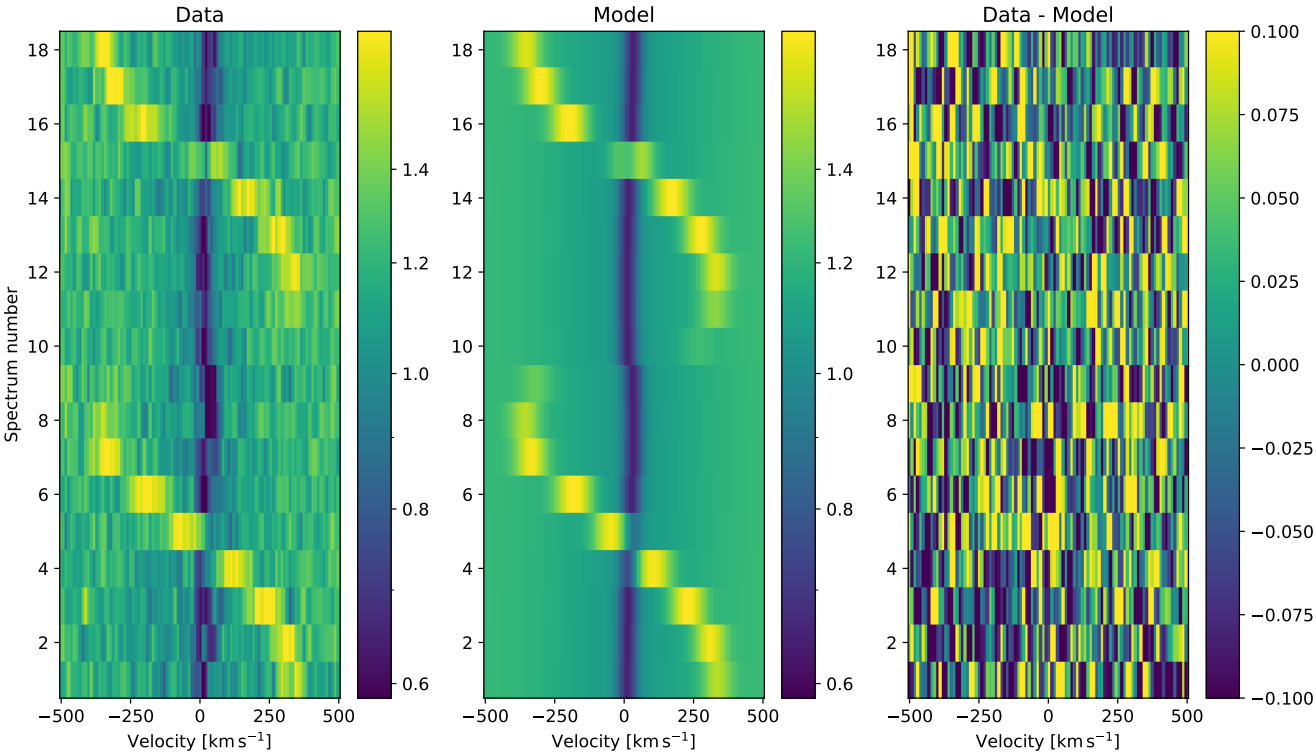

Figure 1. Trailed spectra of the Hα line in ZTF J1828+2308. *Left:* the original data, where the non-LTE core of the white dwarf absorption line is clear as well as emission from the companion moving in anti-phase. The slight velocity jump between spectrum 9 and 10 is due to the fact that the data were taken on different nights and some orbital phases are missed. *Centre:* our best fit model to the data. *Right:* the residuals to the fit.

The emission is strongest around phase 0.5 (roughly spectra 5 and 15 in Figure 1) and weakens towards phase 1 (the eclipse of the white dwarf). Note that there is a small gap in the coverage of our X-shooter data between orbital phases 0.8 and 1.15 (i.e. around the eclipse of the white dwarf), hence the slight jump in the data between spectrum 9 and 10 in Figure 1.

While subtle, we also clearly detect the motion of the white dwarf in the core of the Hα line. Since the higher order Balmer lines lack this strong core, the Hα line is the only feature that we could use to measure the white dwarf's velocity. We therefore focused on measuring the velocities of both stars by fitting the Hα line in all our spectra.

We fitted the Hα line following the method of Parsons et al. (2017b), in which the line is fitted with a combination of a first order polynomial and three Gaussian components. Two Gaussian components with the same velocity are used for the white dwarf absorption line, a wide component for the wings of the line and a narrower component for the core. The third Gaussian component is used for the emission from the companion star. All of the spectra are fitted simultaneously, with the white dwarf absorption components changing position according to $\gamma_{\rm WD} + {\rm K}_{\rm WD} \sin(2\pi\phi)$, where ϕ is the orbital phase (note that this results in $\rm K_{WD}$ having a negative value due to setting $\phi = 0$ as the mid-eclipse of the white dwarf, but only the amplitude is important, so we drop the sign when reporting this value), and the emission component from the companion changing position according to $\gamma_{\rm BD} + {\rm K}_{\rm emis} \sin(2\pi\phi)$. The widths and heights of the two white dwarf components remain the same in all spectra, while the height of the emission component from the companion is modulated as $(1 - \cos\phi)/2$, to simulate the irradiation effect (its width remains the same in all spectra).

We used the ephemeris from Brown et al. (2023) to determine the orbital phase of each spectrum, $\phi = (T_{\rm obs} - T_0)/P_{\rm orb}$, where $T_{\rm obs}$ is the mid-exposure time of the spectrum, T_0 is the time of mid-eclipse of the white dwarf and $P_{\rm orb}$ is the orbital period, but also allowed the values to vary within the quoted uncertainties by placing Gaussian priors on T_0 and $P_{\rm orb}$. The fit was performed using the Markov chain Monte Carlo (MCMC) method as implemented in the EMCEE Python package (Foreman-Mackey et al. 2013). We used 100 walkers, with a burn-in period of 1500, and 5000 production steps.

The best fit model and residuals to the fit are shown in the central and right-hand panels of Figure 1. We found that the radial velocity semi-amplitude of the white dwarf ($\rm K_{WD}$) is extremely small ($10.5 \pm 1.1\,{\rm km\,s^{-1}}$) despite the short period and high inclination of the binary, strongly implying a very low mass for the companion. This is also consistent with the large emission-line velocity for the companion star ($343.4 \pm 1.2\,{\rm km\,s^{-1}}$), although since this emission originates from the inner hemisphere of the companion it does not track the centre of mass of the star, but the centre of light of the emission region. As such it represents a lower limit on the true radial velocity of the companion ($K_{\rm BD}$), which are related via

$$K_{\rm BD} = \frac{K_{\rm emis}}{1 - f(1+q)R_{\rm BD}/a}, \quad (1)$$

where $q = M_{\rm BD}/M_{\rm WD}$ is the mass ratio, $R_{\rm BD}/a$ is the radius of the companion star scaled to the semi-major axis and f is is a constant between 0 and 1 which depends upon the location of the centre of light

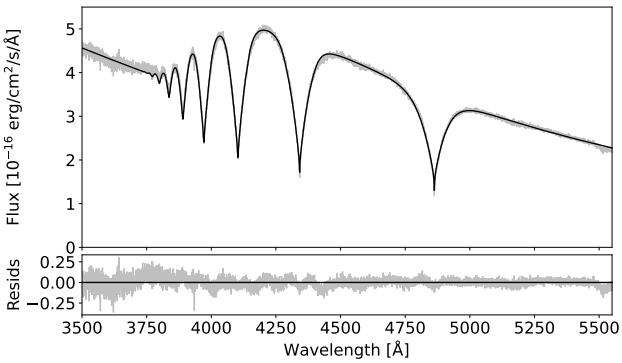

Figure 2. Averaged X-shooter UVB arm spectrum of ZTF J1828+2308, where all spectra have been shifted into the white dwarf frame before averaging. The best fit Koester (2010) model is overplotted in black and the residuals to the fit are shown in the lower panel. Note that the only spectroscopic feature detected from the companion to the white dwarf is weak Hα emission, no other hydrogen lines show detectable emission components.

(Wade & Horne 1988; Parsons et al. 2012a). In the most extreme case ($f = 1$) the emission is assumed to entirely originate at the point on the companion star's surface closest to the white dwarf (the substellar point), while $f = 0$ assumes that the emission is uniform over the surface of the star and hence does track the centre of mass of the star. Using $f = 0$ (i.e. $K_{\rm BD} = K_{\rm emis}$) gives an upper limit to the mass ratio of $q < K_{\rm WD}/K_{\rm emis} = 0.03$, significantly smaller than what is typically found in white dwarf plus brown dwarf binaries ($q \sim 0.1$), confirming the very low mass of the companion star (alternatively, a very high white dwarf mass is possible, although we will see in the next sections that this is not the case).

3.1.2 White dwarf spectral fit

Having calculated the white dwarf's radial velocity semi-amplitude, we shifted all our spectra into the rest frame of the white dwarf and averaged them together to produce a high quality spectrum of the white dwarf. We then fitted this white dwarf spectrum using Koester (2010) models in order to constrain the effective temperature and surface gravity. We also included a polynomial curve in the fit to account for slit losses and the imperfect flux calibration of the X-shooter data. The models were convolved with a Gaussian to match the resolution of the data before fitting. Our short exposure times (and the small velocity of the white dwarf) means that the movement of the white dwarf during each exposure is negligible and hence there is no additional smearing of the spectral features. We do not include any corrections for interstellar reddening.

The average UVB arm spectrum and best fit model for ZTF J1828+2308 are shown in Figure 2. We find $T_{\rm eff} = 15900 \pm 75$ K and $\log(g) = 8.05 \pm 0.05$, where systematic uncertainties have been applied to both values based on Hollands et al. (2024). We found that applying the 3D corrections of Tremblay et al. (2013) made no difference to the parameters of this white dwarf due to its temperature being too high for convection. Assuming the mass-radius relationship from Bédard et al. (2020), this implies a white dwarf mass of around 0.6 M$_\odot$. These values are consistent with those found by Brown et al. (2023) from analysis of multi-band eclipse light curves.

We also detected a weak Ca II 3934 Å absorption line in the spectra of ZTF J1828+2308, but do not detect any velocity variations from this line. It is also very narrow leading us to conclude that this is most likely an interstellar absorption feature. No other lines except the hydrogen Balmer series are detected in the X-shooter spectrum of ZTF J1828+2308.

3.1.3 Light curve fit

Our high-speed, multi-band HiPERCAM light curves of the eclipse of the white dwarf in ZTF J1828+2308 show an extremely deep eclipse in all bands (no flux is detected from the companion during the eclipse in any band, see Figure 3). The shape of the eclipse is sensitive to the sizes of the stars as well as the inclination of the binary. However, in general there is not sufficient information in the eclipse light curve alone to simultaneously solve for the radii of both stars and also the orbital inclination (Parsons et al. 2017a). This can be overcome by also including any additional constraints from the spectroscopic analysis when fitting the light curves. Of particular importance for this system is the measurement of the surface gravity of the white dwarf from the spectral fit and the constraints on the radial velocities of the two stars. This is because knowledge of both radial velocities yields the masses for any given inclination (i) via Kepler's third law,

$$\frac{P_{\rm orb} K_{1,2}^3}{2\pi G} = \frac{M_{2,1} \sin^3 i}{(1+q)^2}, \tag{2}$$

where the subscripts 1 and 2 refer to the two stars (i.e. K_1 is used with M_2 and vice versa) and $q = K_1/K_2$. At the same time, the shape of the eclipse yields the radii of the two stars for a given inclination, so combining these two pieces of information means that we can determine the surface gravity of the white dwarf at any inclination. We can then use the spectroscopically determined value of $\log g$ to determine the correct inclination and hence stellar and binary parameters.

Unfortunately we do not have a direct measurement of the radial velocity of the companion to the white dwarf, but a lower limit from the Hα emission line. We can use Equation 1 to correct the emission line velocity to the centre of mass velocity, but a value of f must be assumed (for a given inclination $R_{\rm BD}/a$ is known from the eclipse shape and since $q = K_{\rm WD}/K_{\rm BD}$ it is possible to solve for this analytically). Therefore, in order to fit the eclipse and determine the stellar and binary parameters a value of f must be chosen. Typically a value around 0.5 is used (e.g. Parsons et al. 2017a) which assumes that the emission is mostly uniformly spread over the heated face of the star. This is also what is measured in systems where there are both irradiation-driven emission lines and absorption lines from the secondary that track the centre of mass (Parsons et al. 2012b). These studies find that different emission lines need different correction factors, based on the optical depth of the emission, with Hα emission typically needing $f = 0.5$ (Parsons et al. 2010), although see van Roestel et al. (2017) for an alternative approach to determining this value. Therefore, we adopt a value of $f = 0.5$ when fitting the light curves of ZTF J1828+2308 (although see the end of this section for an alternative approach).

With a value chosen for the f term in Equation 1 we were then able to fit the eclipse light curves to determine the stellar and binary parameters. The fit was performed using LCURVE (Copperwheat et al. 2010), a code specifically designed for fitting the light curves of white dwarf binaries, which accounts for Roche distortion and irradiation. We fitted for the two masses, $M_{\rm WD}$ and $M_{\rm BD}$, the two radii, $R_{\rm WD}$ and $R_{\rm BD}$, the effective temperature of the white dwarf, $T_{\rm eff,WD}$, the orbital inclination, i, and the time of mid-eclipse, T_0. We held the orbital period fixed at the value from Brown et al. (2023).

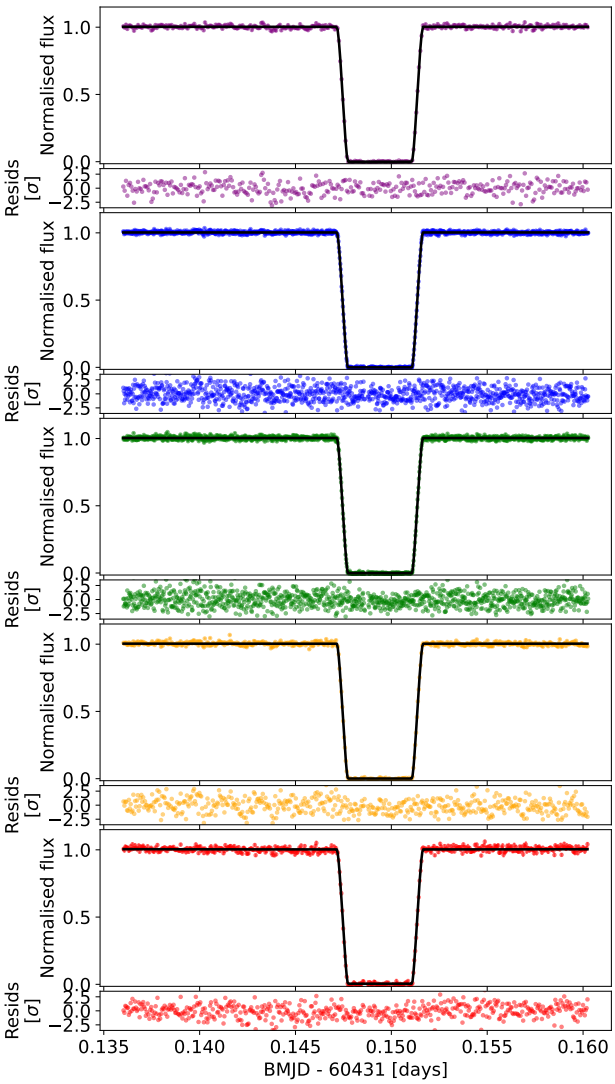

Figure 3. GTC+HiPERCAM eclipse light curves of ZTF J1828+2308 (top to bottom: u_s, g_s, r_s, i_s and z_s bands) along with the best fit models (black lines). Residuals to the fits are shown beneath each panel (in standard deviations).

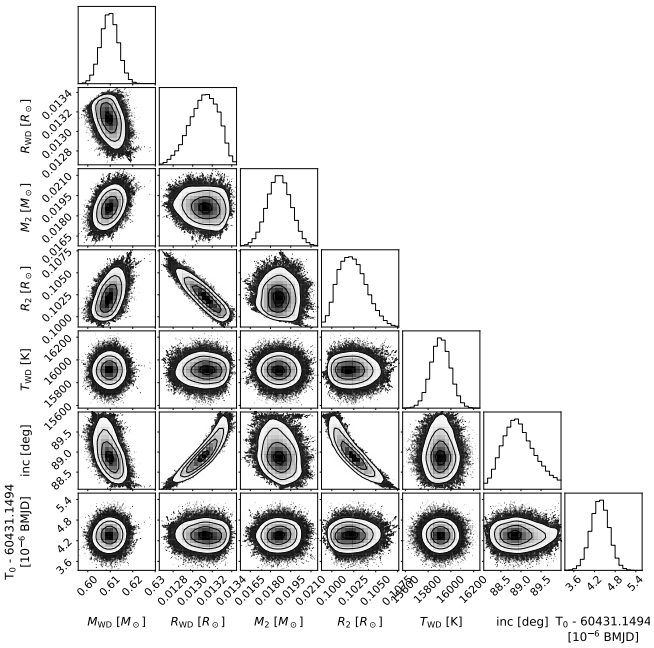

Figure 4. Posterior probability distributions for model parameters obtained through fitting the GTC+HiPERCAM light curves of ZTF J1828+2308.

LCURVE requires as input: the mass ratio, the orbital inclination, the radii of both stars scaled by the separation (the separation can be calculated from the masses and orbital period), the temperatures of the two stars (we fix the temperature of the companion to 1500 K, but since it is undetected in the light curves this has no effect on the fit), the limb-darkening coefficients for both stars, which we set for the white dwarf by calculating its $\log g$ value from the mass and radius and interpolating the 4-parameter non-linear limb darkening values from Claret et al. (2020) in each band. The limb darkening coefficients of the companion star have no effect on the fit, so we set these to a simple linear coefficient of 0.5 in all bands. LCURVE also requires a central eclipse time and an orbital period but all other parameters are irrelevant for this fit.

The light curves are only weakly dependent upon the temperature of the white dwarf. In principle the out-of-eclipse flux levels in each band constrain the temperature, but the accuracy of this is dependent on the quality of the flux calibration. Moreover, LCURVE computes models assuming black body temperatures, so a conversion must be made (e.g. Brown et al. 2022). Given that we already had a precise measurement for the white dwarf temperature from fitting the spectrum, we decided to normalise each of our light curves before fitting and constrain the white dwarf temperature via a Gaussian prior on the fit based on the spectroscopic value. This means that a conversion to a black body temperature is not required and also has the advantage that the temperature can still vary during the fit within the uncertainties from the spectroscopic fit, allowing a full exploration of the range of limb darkening coefficients for the white dwarf (which depend on its temperature). Given the extremely well sampled and high signal-to-noise ratio of our HiPERCAM light curves, the fit is sensitive to the limb darkening of the white dwarf and so we did not want to fix these values. The consequence of this is that the white dwarf temperature value (and uncertainty) returned by the fit is essentially just the spectroscopic value and so we do not improve on this with the light curve data.

Along with the prior on the white dwarf temperature, we also placed a Gaussian prior on the surface gravity of the white dwarf, based on the spectroscopically measured value. Finally, for each model we also computed the expected values of $K_{\rm WD}$ and $K_{\rm BD}$ using Equation 2 and hence $K_{\rm emis}$ (based on Equation 1 with $f = 0.5$). We placed Gaussian priors on $K_{\rm WD}$ and $K_{\rm emis}$ based on the measured values and their uncertainties. We then performed an MCMC fit of the light curves in all band simultaneously using EMCEE. We used 100 walkers, with a burn-in period of 2000 and 10000 production steps.

The HiPERCAM eclipse light curves along with the best fit model are shown in Figure 3, the best fit parameters are listed in Table 2 and the MCMC parameter distributions are shown in Figure 4. We also calculated the white dwarf cooling age from the Bédard et al. (2020) models and the volume-averaged Roche lobe filling factor of the brown dwarf, both of which are listed in Table 2. We find